

## 4.6 Plane duality

In this section we shall use MacLane's theorem to uncover another connection between planarity and algebraic structure: a connection between the duality of plane graphs, defined below, and the duality of the cycle and cut space hinted at in Chapters 1.9 and 2.4.

plane  
multigraph

A *plane multigraph* is a pair  $G = (V, E)$  of finite sets (of *vertices* and *edges*, respectively) satisfying the following conditions:

- (i)  $V \subseteq \mathbb{R}^2$ ;
- (ii) every edge is either an arc between two vertices or a polygon containing exactly one vertex (its *endpoint*);
- (iii) apart from its own endpoint(s), an edge contains no vertex and no point of any other edge.

We shall use terms defined for plane graphs freely for plane multigraphs. Note that, as in abstract multigraphs, both loops and double edges count as cycles.

Let us consider the plane multigraph  $G$  shown in Figure 4.6.1. Let us place a new vertex inside each face of  $G$  and link these new vertices up to form another plane multigraph  $G^*$ , as follows: for every edge  $e$  of  $G$  we link the two new vertices in the faces incident with  $e$  by an edge  $e^*$  crossing  $e$ ; if  $e$  is incident with only one face, we attach a loop  $e^*$  to the new vertex in that face, again crossing the edge  $e$ . The plane multigraph  $G^*$  formed in this way is then dual to  $G$  in the following sense: if we apply the same procedure as above to  $G^*$ , we obtain a plane multigraph very similar to  $G$ ; in fact,  $G$  itself may be reobtained from  $G^*$  in this way.

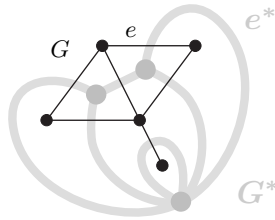


Fig. 4.6.1. A plane graph and its dual

To make this idea more precise, let  $G = (V, E)$  and  $(V^*, E^*)$  be any two plane multigraphs, and put  $F(G) =: F$  and  $F((V^*, E^*)) =: F^*$ . We call  $(V^*, E^*)$  a *plane dual* of  $G$ , and write  $(V^*, E^*) =: G^*$ , if there are bijections

plane dual  
 $G^*$

$$\begin{array}{lll} F \rightarrow V^* & E \rightarrow E^* & V \rightarrow F^* \\ f \mapsto v^*(f) & e \mapsto e^* & v \mapsto f^*(v) \end{array}$$

satisfying the following conditions:

- (i)  $v^*(f) \in f$  for all  $f \in F$ ;
- (ii)  $|e^* \cap G| = |\dot{e}^* \cap \dot{e}| = |e \cap G^*| = 1$  for all  $e \in E$ , and in each of  $e$  and  $e^*$  this point is an inner point of a straight line segment;
- (iii)  $v \in f^*(v)$  for all  $v \in V$ .

Every connected plane multigraph has a plane dual. Indeed, to satisfy condition (i) we start by picking from each face  $f$  of  $G$  a point  $v^*(f)$  as a vertex for  $G^*$ . We can then link these vertices up by independent arcs as required by (ii), and using the connectedness of  $G$  show that there is indeed a bijection  $V \rightarrow F^*$  satisfying (iii) (Exercise 37).

If  $G_1^*$  and  $G_2^*$  are two plane duals of  $G$ , then clearly  $G_1^* \cong G_2^*$ ; in fact, one can show that the natural bijection  $v_1^*(f) \mapsto v_2^*(f)$  is a topological isomorphism between  $G_1^*$  and  $G_2^*$ . In this sense, we may speak of *the* plane dual  $G^*$  of  $G$ .

Finally,  $G$  is in turn a plane dual of  $G^*$ . Indeed, this is witnessed by the inverse maps of the bijections from the definition of  $G^*$ : setting  $v^*(f^*(v)) := v$  and  $f^*(v^*(f)) := f$  for  $f^*(v) \in F^*$  and  $v^*(f) \in V^*$ , we see that conditions (i) and (iii) for  $G^*$  transform into (iii) and (i) for  $G$ , while condition (ii) is symmetrical in  $G$  and  $G^*$ . As duals are easily seen to be connected (Exercise 36), this symmetry implies that connectedness is also a necessary condition for  $G$  to have a dual.

Perhaps the most interesting aspect of plane duality is that it relates geometrically two types of edge sets – cycles and bonds – that we have previously seen to be algebraically related (Theorem 1.9.4):

**Proposition 4.6.1.** *For any connected plane multigraph  $G$ , an edge set  $E \subseteq E(G)$  is the edge set of a cycle in  $G$  if and only if  $E^* := \{e^* \mid e \in E\}$  is a bond in  $G^*$ .*

[6.5.2]

*Proof.* By conditions (i) and (ii) in the definition of  $G^*$ , two vertices  $v^*(f_1)$  and  $v^*(f_2)$  of  $G^*$  lie in the same component of  $G^* - E^*$  if and only if  $f_1$  and  $f_2$  lie in the same region of  $\mathbb{R}^2 \setminus \bigcup E$ : every  $v^*(f_1) - v^*(f_2)$  path in  $G^* - E^*$  is an arc between  $f_1$  and  $f_2$  in  $\mathbb{R}^2 \setminus \bigcup E$ , and conversely every such arc  $P$  (with  $P \cap V(G) = \emptyset$ ) defines a walk in  $G^* - E^*$  between  $v^*(f_1)$  and  $v^*(f_2)$ .

(4.1.1)  
(4.2.4)

Now if  $C \subseteq G$  is a cycle and  $E = E(C)$  then, by the Jordan curve theorem and the above correspondence,  $G^* - E^*$  has exactly two components. So  $E^*$  is a bond of  $G^*$ , a minimal non-empty cut.

Conversely, if  $E \subseteq E(G)$  is such that  $E^*$  is a cut in  $G^*$  then, by Proposition 4.2.4 and the above correspondence,  $E$  contains the edges of a cycle  $C \subseteq G$ . If  $E^*$  is a bond,  $E$  cannot contain any further edges (by the implication shown before). Hence  $E = E(C)$ .  $\square$

abstract  
dual

Proposition 4.6.1 suggests the following generalization of plane duality to abstract multigraphs.<sup>5</sup> Call a multigraph  $G^*$  an *abstract dual* of a multigraph  $G$  if  $E(G^*) = E(G)$  and the bonds in  $G^*$  are precisely the edge sets of cycles in  $G$ . (Neither  $G$  nor  $G^*$  need be connected now.)

This correspondence between cycles and bonds extends to the spaces they generate:

**Proposition 4.6.2.** *If  $G^*$  is an abstract dual of  $G$ , then the cut space of  $G^*$  is the cycle space of  $G$ , i.e.,*

$$\mathcal{B}(G^*) = \mathcal{C}(G).$$

(1.9.3) *Proof.* Since the cycles of  $G$  are precisely the bonds of  $G^*$ , the subspace  $\mathcal{C}(G)$  they generate in  $\mathcal{E}(G) = \mathcal{E}(G^*)$  is the same as the subspace generated by the bonds in  $G^*$ . By Lemma 1.9.3, this is the space  $\mathcal{B}(G^*)$ .  $\square$

(1.9.4) By Theorem 1.9.4, Proposition 4.6.2 implies at once that if  $G^*$  is an abstract dual of  $G$  then  $G$  is an abstract dual of  $G^*$ . One can show that if  $G$  is 3-connected, then  $G^*$  is unique (up to isomorphism and the addition of isolated vertices). By Lemma 3.1.3, a non-empty subset of  $E(G) = E(G^*)$  is the edge set of a block of  $G$  if and only if it is the edge set of a block of  $G^*$ . By Lemma 3.1.2, this implies that the blocks of  $G^*$  are duals of the blocks of  $G$ .

(3.1.3)  
(3.1.2)

Although the notion of abstract duality arose as a generalization of plane duality, it could have been otherwise. We knew already from Theorem 1.9.4 that the cycles and the bonds of a graph form natural and related sets of edges. It would not have been unthinkable to ask whether, for some graphs, the orthogonality between these collections of edge sets might give them sufficiently similar intersection patterns that a collection forming the cycles in one graph could form the bonds in another, and vice versa. In other words, for which graphs can we move their entire edge set to a new set of vertices, redefining incidences, so that precisely those sets of edges that used to form cycles now become bonds (and vice versa)? Put in this way, it seems surprising that this could ever be achieved, let alone for such a large and natural class of graphs as all planar graphs.

As one of the highlights of classical planarity theory we now show that the planar graphs are *precisely* those for which this can be done. Admitting an abstract dual thus appears as a new planarity criterion. Conversely, the theorem can be read as a surprising topological characterization of the equally fundamental property of admitting an abstract dual:

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<sup>5</sup> In what follows we shall use some lemmas from earlier chapters that were stated for graphs only. These lemmas extend to multigraphs with proofs unchanged.

**Theorem 4.6.3.** (Whitney 1932)

*A graph is planar if and only if it has an abstract dual.*

*Proof.* Let  $G$  be a planar graph, and consider any drawing. Every component  $C$  of this drawing has a plane dual  $C^*$ . Consider these  $C^*$  as abstract multigraphs, and let  $G^*$  be their disjoint union. Then the bonds of  $G^*$  are precisely those of the  $C^*$ , which by Proposition 4.6.1 correspond to the cycles in  $G$ .

(1.9.2)  
(4.5.1)

Conversely, suppose that  $G$  has an abstract dual  $G^*$ . For a proof that  $G$  is planar, it suffices by Theorem 4.5.1 and Proposition 4.6.2 to show that  $\mathcal{B}(G^*)$  has a sparse basis. By Proposition 1.9.2, it does.  $\square$

The duality theory for both abstract and plane graphs can be extended to infinite graphs. As these can have infinite bonds, their duals must then have ‘infinite cycles’. Such things do indeed exist, and are fascinating: they arise as topological circles in a space formed by the graph and its *ends*; see Chapter 8.6.

## Exercises

1. Show that every graph can be embedded in  $\mathbb{R}^3$  with all edges straight.
2. <sup>-</sup> Show directly by Lemma 4.1.2 that  $K_{3,3}$  is not planar.
3. Here is an inductive ‘proof’ that every maximal plane graph of order  $\geq 4$  is a plane triangulation of minimum degree 3. The induction starts with  $K^4$ . For the induction step, consider an arbitrary maximal plane graph  $G$  of order  $n \geq 4$ , and consider all possible ways of extending it to a maximal plane graph  $G'$  of order  $n+1$  by adding a new vertex  $v$ . No matter how this is done,  $v$  will come to sit in a face of  $G$ , which by the inductive assumption is bounded by a triangle. Since  $G'$  is maximally planar,  $v$  must be joined to all three vertices of that triangle. Clearly,  $G'$  is another plane triangulation, and  $\delta(G') = d(v) = 3$ .
  - (i) <sup>-</sup> Find the flaw in this ‘proof’.
  - (ii) Find a counterexample, and explain why the ‘proof’ overlooks it.
4. Show that every planar graph is a union of three forests.
5. The ancient Greeks loved regular plane graphs whose faces were bounded by cycles of the same length.
  - (i) Show that such graphs exist for only finitely many pairs  $(d, \ell)$  of degree  $d \geq 3$  and cycle length  $\ell$ . Can you give an upper bound?
  - (ii) <sup>+</sup> Show that there are only finitely many such plane graphs, up to topological isomorphism.
6. A *fullerene* is a molecule that is made up entirely of carbon atoms forming a cubic plane graph all whose faces are pentagons or hexagons. Show that, since carbon atoms can form double bonds, every such graph can be realized in principle by (4-valent) carbon atoms.